

Ibn Tofail University

Analysis IV

Problem Set I

Exercise 1:

Study the convergence of the following series:

1. $\sum_{n=1}^{+\infty} \frac{1}{n}$

2. $\sum_{n=2}^{+\infty} \frac{n^2 + 1}{n^2}$

3. $\sum_{n=1}^{+\infty} \frac{2}{\sqrt{n}}$

4. $\sum_{n=1}^{+\infty} \frac{(2n+1)^4}{(7n^2+1)^3}$

5. $\sum_{n=1}^{+\infty} \left(ne^{\frac{1}{n}} - n \right)$

6. $\sum_{n=0}^{+\infty} \ln(1 + e^{-n})$

7. $\sum_{n=1}^{+\infty} \left(\frac{n}{n+1} \right)^{n^2}$

Correction

Solution here.

Exercise 2:

Determine the nature of the numerical series with the following general terms:

1. $u_n = \frac{1}{n \cos^2 n}$

2. $v_n = \frac{1}{[\ln(n)]^n}$

3. $w_n = \frac{\tan^n\left(\frac{\pi}{7}\right)}{3^{n+2}}$

4. $x_n = \frac{1}{n \ln(n)}$

5. $y_n = \frac{1}{n(\ln n)^2}$

6. $z_n = \frac{1}{(\ln 2)^2 + (\ln 3)^2 + \dots + (\ln n)^2}$

Correction

Solution here.

Exercise 3:

Show that the series with general term

$$w_n = \frac{(-1)^n}{\ln(1 + \sqrt{n})}$$

is convergent but not absolutely convergent (called semi-convergent).

Correction

Solution here.

Exercise 4:

1. Show that the following two series are absolutely convergent:

$$\sum_{n=0}^{+\infty} \frac{1}{n!} \quad \text{and} \quad \sum_{n=0}^{+\infty} \frac{(-1)^n}{2^n}$$

2. Deduce that the series:

$$\sum_{n=0}^{+\infty} \left[\sum_{k=0}^n \frac{(-1)^{n-k}}{k! \cdot 2^{n-k}} \right]$$

is absolutely convergent and determine its value.

Correction

Solution here.

Exercise 5:

Study the nature of the following series:

$$\sum_{n>2} \ln \left(1 + \frac{(-1)^n}{n} \right), \quad \sum_{n>1} \sin \left(\frac{(-1)^n}{n} \right)$$

Correction

Solution here.

Exercise 6:

Show that the series with general terms

$$u_n := \frac{(-1)^n}{\sqrt{n}} \quad \text{and} \quad v_n := \frac{(-1)^n}{\sqrt{n + (-1)^n}}$$

are not of the same nature, although $u_n \sim v_n$.

Correction

Solution here.